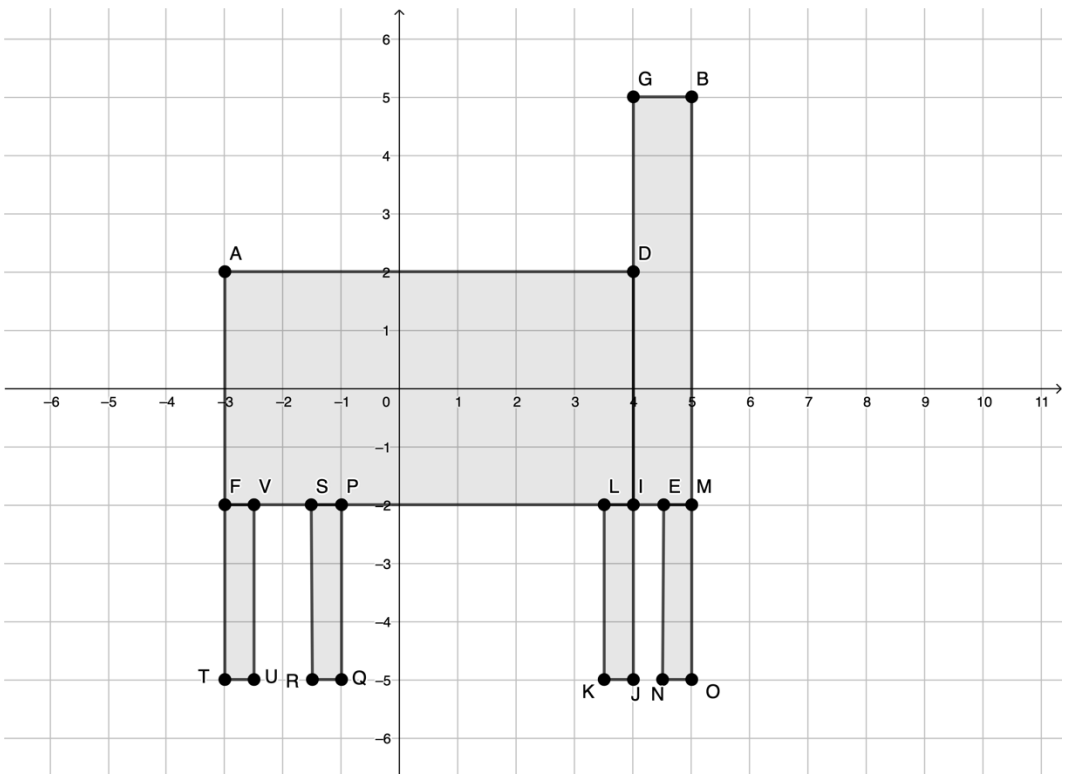


1. Brandon is creating a layout for his new arcade game. Each part of the model is made of quadrilaterals. The front legs, $LIJK$ & $EMON$, and the back legs, $FVUT$ & $SPQR$, are all congruent rectangles. $GBMI$ and $ADIF$ are rectangles.

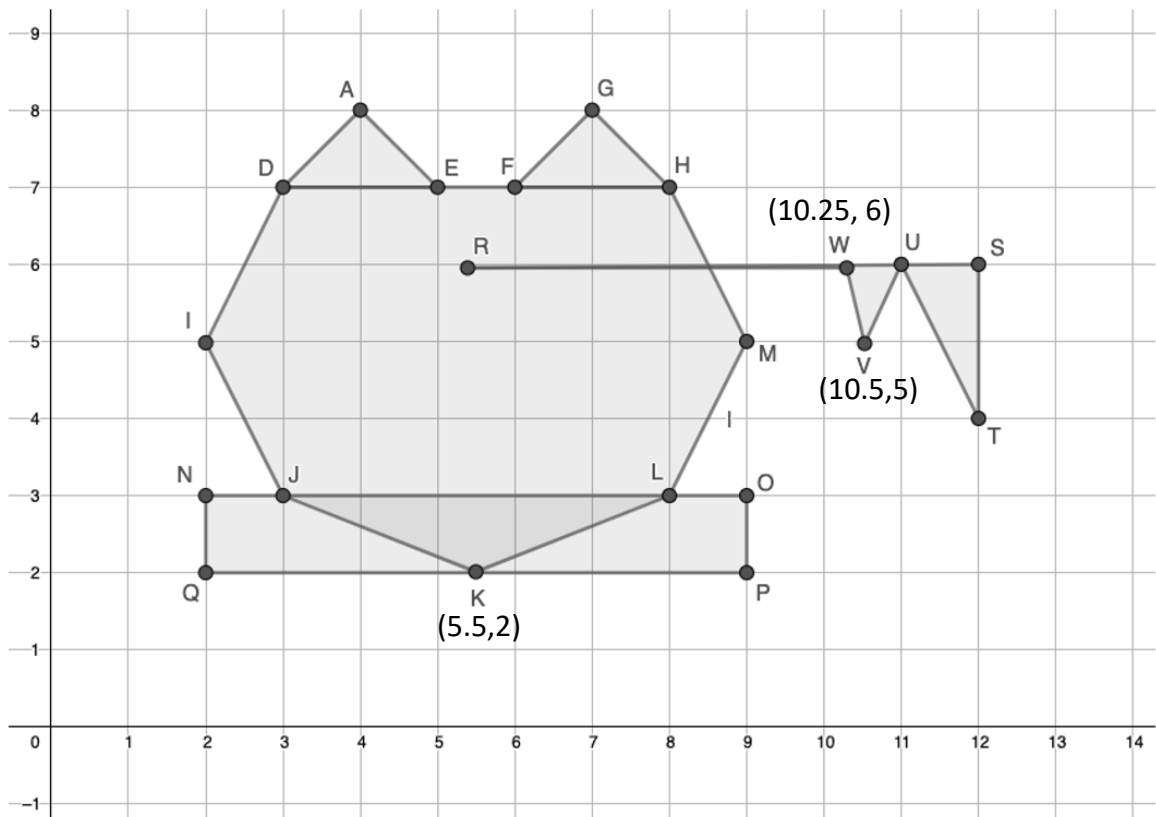


- a. Complete the table. If necessary, round to the nearest thousandth.

Section	Perimeter	Area
Front legs $LIJK, EMON$	7 each, 14 total	1.5 each, 3 total
Back legs $FVUT, SPQR$	7 each, 14 total	1.5 each, 3 total
Rectangle $GBMI$	16	7
Rectangle $ADIF$	22	28

- b. What is the perimeter of the composite figure made by all of the sections?
 If necessary, round to the nearest thousandth.
 $66 - 8(0.5) - 2(4) = 54$ units
- c. What is the area of the composite figure made by all of the sections?
 If necessary, round to the nearest thousandth.
 41 square units
- d. If each square unit weights 20 lbs. How much will the arcade game weigh?
 $41(20) = 820$ lbs.

2. Asia is building her very first Battle Bot. She makes her first sketch of her battle bot's profile on the grid. Each square on the grid represents 1 inch.



- a. Complete the table. If necessary, round to the nearest thousandth.

Section	Area (in ²)	Perimeter (in)
Tail cover <i>NQPO</i>	7	16
Triangles <i>ADE</i> and <i>GFH</i>	1 each, 2 total	≈ 4.828 each, ≈ 9.567 total
Triangle <i>STU</i>	1	≈ 5.236
Triangle <i>WUV</i>	0.375	0.75 + 1.031 + 1.118 = 2.899
Battle Bot Body <i>DIJKLMH</i>	26.5	≈ 19.329

- b. If every unit square is equal to 2.5 pounds, what is the total weight of the Battle Bot?
 36.875(2.5) ≈ 92.188 lbs.